
Q1. Define a relation T on Z as follows: For all integers x and y (8 marks)

$$x T y \Leftrightarrow 4 \mid (3x + y)$$

a. Is $1 T 1$?

b. Is $2 T (-2)$?

c. Is $5 T 3$?

d. Is $(-1) T 11$?

Q2. Let $A = \{0, 1, 2, 3\}$. For each relation below:

- 1. Draw the directed graph. (5 x 3 = 15)**
- 2. Determine if it is Reflexive, Symmetric, or Transitive. (3 x 3 = 9)**
- 3. Provide counterexamples for any properties that fail. (3)**

1. $R_1 = \{(0, 0), (1, 1), (2, 2), (3, 3), (0, 2), (2, 0)\}$

2. $R_2 = \{(0, 1), (1, 0), (1, 2), (2, 1)\}$

3. $R_3 = \{(0, 0), (1, 1), (0, 1), (1, 0)\}$